



Department of Civil Engineering
Indian Institute of Technology, Bombay, India

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A credit seminar presentation on

Finite Element Modelling of Bolted Joints under Static and Dynamic Loads

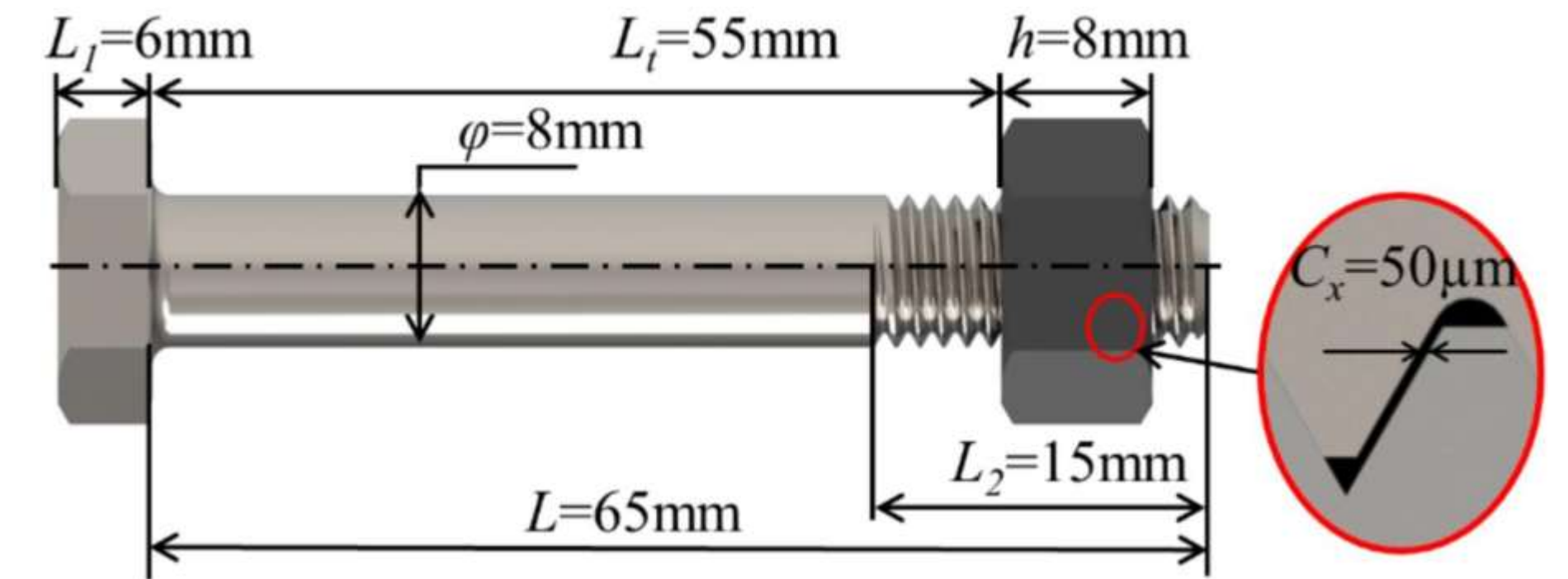
By

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Motivation

- Threaded fasteners are essential in critical structures (e.g., aerospace).
- Two main failure modes:
 - Fatigue failure from axial loads.
 - Loosening failure from transverse vibrations.
- Existing fatigue improvement methods (e.g., new materials, surface treatments) increase cost and doesn't improve anti-loosening.
- Existing anti-loosening methods (e.g., interference fit, friction enhancement) increases friction but it worsens fatigue life due to stress concentration.
- Hence, it is difficult to optimize both fatigue and loosening resistance simultaneously.
- This study introduces a stress-guided design method.
- Uses geometric tuning (pitch difference α and interference volume δ) instead of expensive materials.
- Achieves simultaneous improvement in fatigue resistance and anti-loosening.



Frictional Force between threads

$$f = F^n \times \mu_s$$

f = Frictional Force between threads

F^n = Thread contact force

μ_s = Friction coefficient of the thread surface

Loosening

Expression for ordinary fasteners

$$y_{R1} = A_1N + A_2N^2 + A_3N^3 + A_4N^4 + A_5N^5 + C_1$$

y_{R1} = Expression for ordinary fastener

N = Thread turn number

$A_1.....A_5$ = Polynomial coefficients

C_1 = Constant

- y_{R1} is predicted σ_n values from the fitted polynomial for ordinary fastener.
- Above expression describes variation of σ_n across thread turns.
- Fifth degree polynomial was obtained using curve fitting of the plot that was plotted by using numerical values of σ_n from FEA simulation under vibration

Expression for special fasteners

$$y_{R1} = A_1N + A_2N^2 + A_3N^3 + A_4N^4 + A_5N^5 + C_1$$

y_{R2} = Expression for special fastener

N = Thread turn number

$B_1.....B_5$ = Polynomial coefficients

C_1 = Constant

- y_{R2} is predicted σ_n values from the fitted polynomial for ordinary fastener.
- Above expression describes variation of σ_n (normal compressive/anti-loosening stress) across thread turns.
- Fifth degree polynomial was obtained using curve fitting of the plot that was plotted by using numerical values of σ_n from FEA simulation under vibration

Formulation

$$f_1 = \left| \int_0^7 y_{R1} \right| = \left| \frac{A_1 N^2}{2} + \frac{A_2 N^3}{3} + \frac{A_3 N^4}{4} + \frac{A_4 N^5}{5} + \frac{A_5 N^6}{6} + C_1 N \right| = 668.82$$

$$f_2 = \left| \int_0^7 y_{R2} \right| = \left| \frac{B_1 N^2}{2} + \frac{B_2 N^3}{3} + \frac{B_3 N^4}{4} + \frac{B_4 N^5}{5} + \frac{B_5 N^6}{6} + C_2 N \right| = 910.28$$

$A_1 = -272.3$, $A_2 = 114.7$, $A_3 = -20$, $A_4 = 2.2$, $A_5 = -0.12$, $C_1 = -42.5$

$B_1 = -354$, $B_2 = 258.8$, $B_3 = -78.6$, $B_4 = 10.6$, $B_5 = -0.5$, $C_2 = -105.1$

f_1 = Area under σ_n (compressive stress) curve for ordinary fastener

f_2 = Area under σ_n curve for special (pitch-difference) fastener ($\alpha = 8 \mu\text{m}$)

Limits from 0-7 as there are 6.4 threads of nut

Expression for equivalent friction coefficient

$$\mu_{eq} = \frac{f_2}{f_1} \mu = 1.36\mu$$

μ_{eq} = Equivalent friction coefficient

μ = Static friction coefficient

μ_{eq} shows how effective the fastener is at using the available μ through geometry

At $\alpha=8^\circ$ fasteners can generate 1.36 times the thread friction torque of ordinary fasteners during vibration

High cycle fatigue test

1. Used to determine number of loading cycles a component can withstand under repeated stress before it fails — when the stress levels are relatively low, but cycles are very high ($>10,000$ cycles)
2. Purpose: measure fatigue life (N_f)
3. According to the ISO 1099 standard for high-cycle fatigue
A mean axial stress is applied
A sinusoidal alternating stress is applied on top
4. Cycles continue until
The fastener cracks, or
The clamping force drops to zero

Formulation

High cycle fatigue test

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

$$\sigma_m = \sigma_{\max} - \sigma_a$$

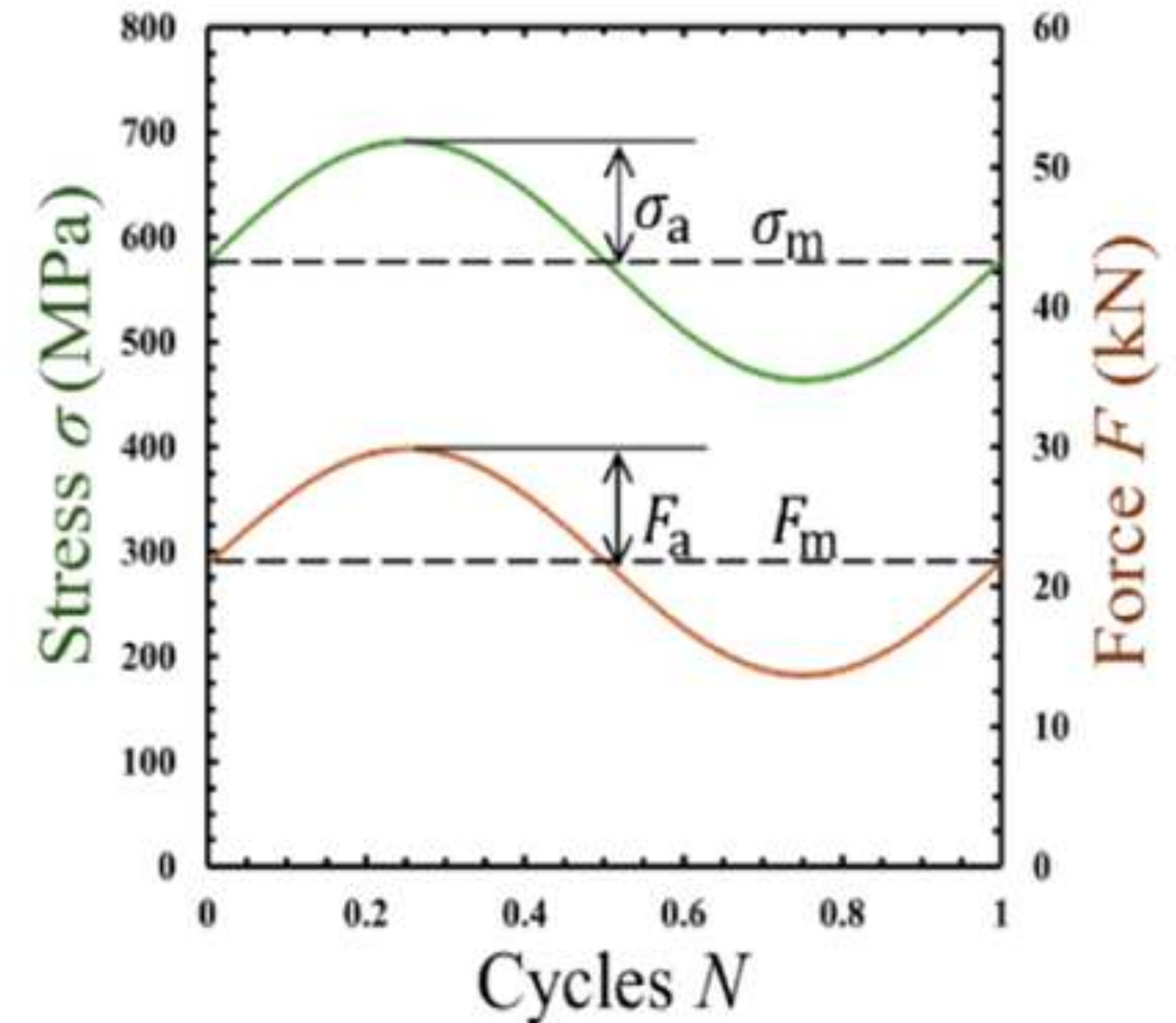
σ_m = Mean stress

σ_a = Stress amplitude (magnitude of alternating stress that goes above and below the mean stress (σ_m))

σ_s = Yield strength

$$\sigma_m = 50\% \sigma_s$$

$$\sigma_a = 20\% \sigma_s, 16.6\% \sigma_s, 13.3\% \sigma_s, 10\% \sigma_s$$



Fatigue tests by an ordinary test bar

$$\frac{\Delta L_t}{L_t} = \varepsilon$$

$$\frac{\sigma_y}{E} = \varepsilon$$

$$\sigma_y = \frac{F}{S}$$

$$S = \pi \cdot \left(\frac{D}{2} \right)^2$$

L_t = original clamped height

ε = Strain

σ_y = Yield strength

E = Young's modulus

F = Clamping Force

S = Area of cross-section

D = Diameter of Bolt

Fatigue Life

$$N_f = n$$

When,

$$L_{t_{new}} = L_t + 2\Delta L_t$$

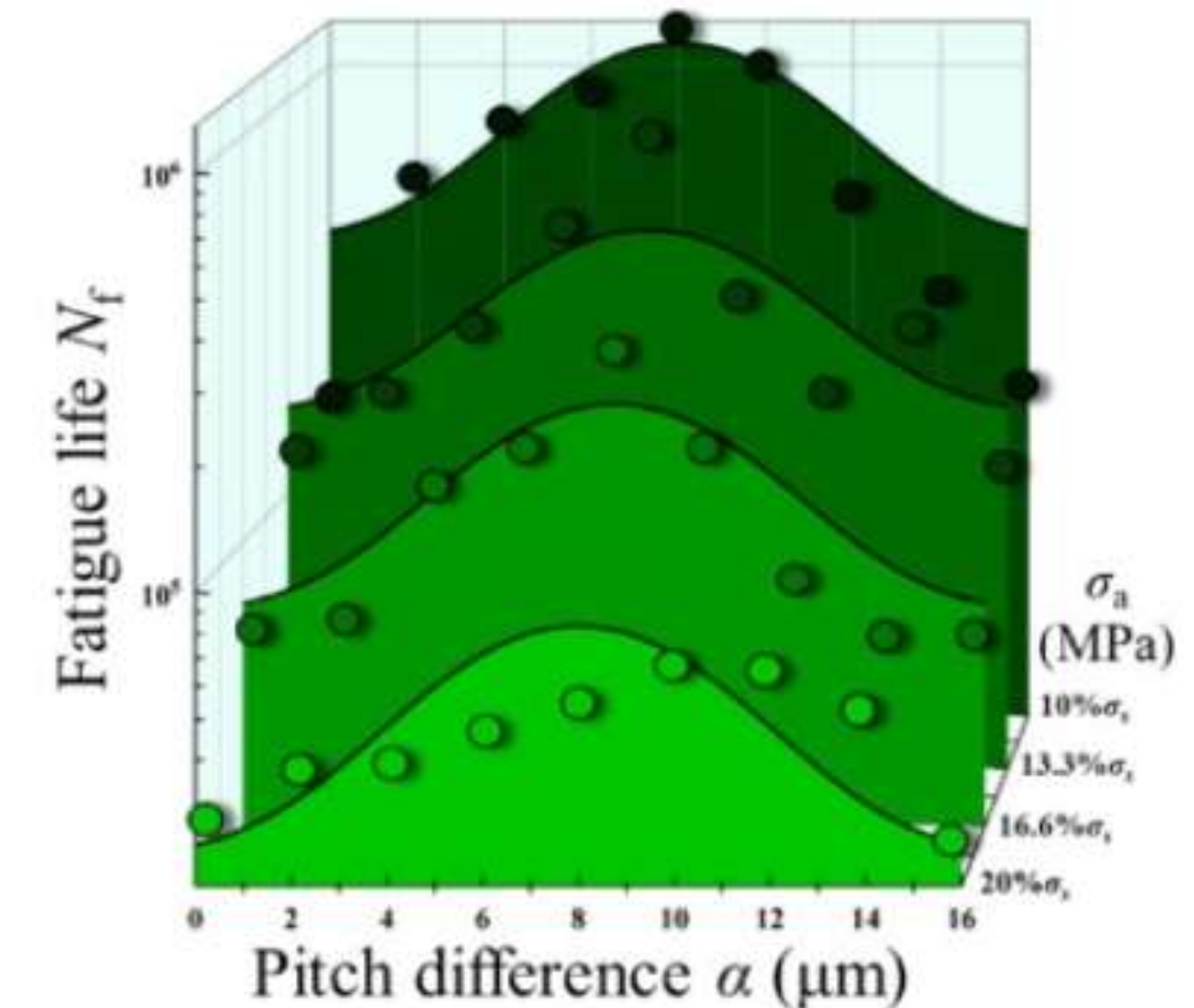
N_f = Fatigue life

L_t = Original clamped height of fastener

n = Number of loading cycles

N_f is n (number of loading cycles) when the fastener stretches twice as much as expected

- When α increases from 0 μm to 8 μm : N_f under each amplitude increases.
- When increases from 8 μm to 16 μm : N_f at each amplitude decreases.
- The fastener with $\alpha=8$ μm has the longest N_f value



(c) α – fatigue life at various amplitudes.

Stüssi S–N Curve

$$\Delta\sigma = \frac{R_m + \gamma N^\beta \Delta\alpha_\infty}{1 + \gamma N^\beta}$$

$$\Delta\sigma = \sigma_{\max} - \sigma_{\min}$$

$\Delta\sigma$: Stress range during the fatigue test

N: number of load cycles up to failure or up to end of the test

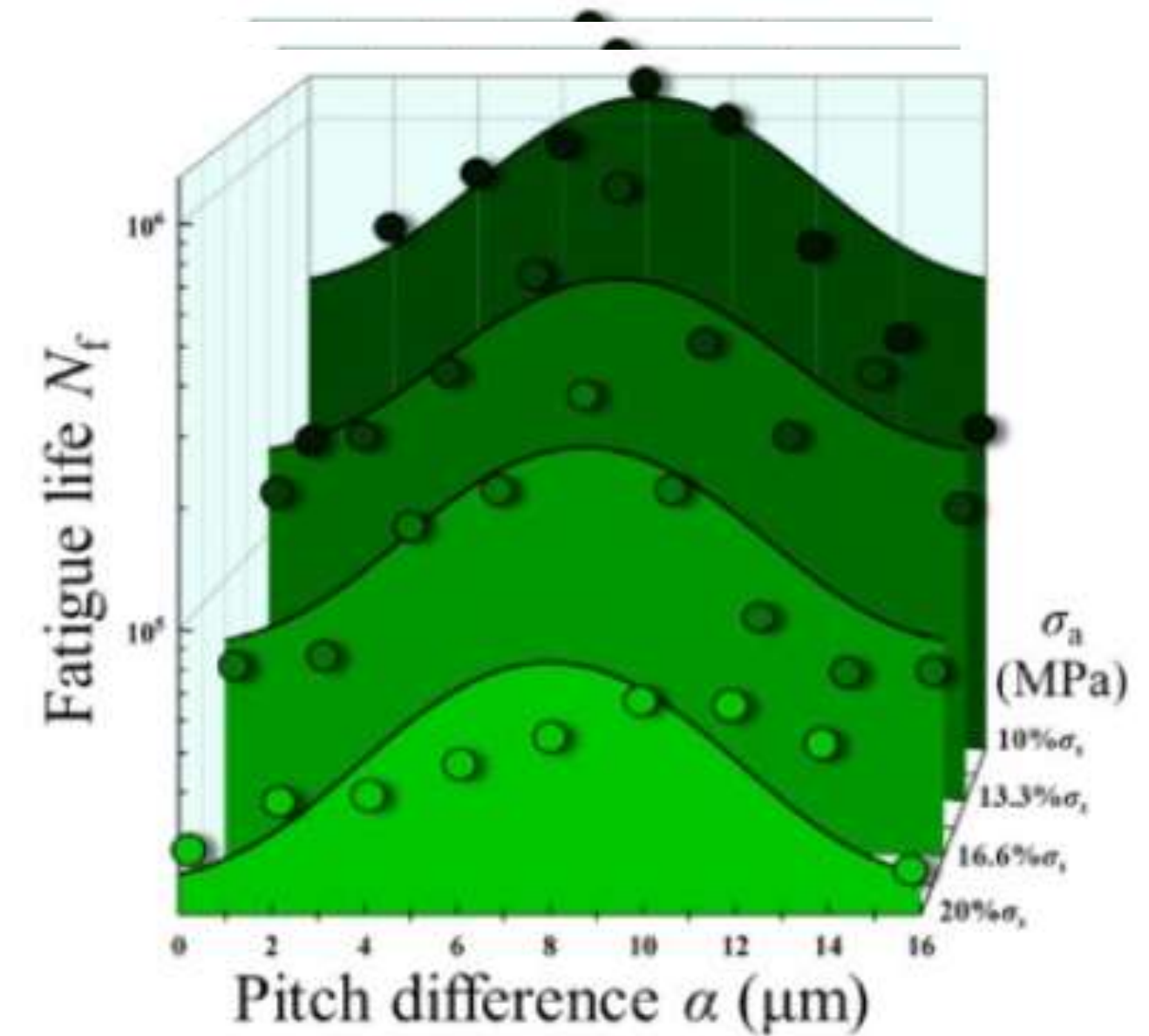
R_m : ultimate tensile strength

$\Delta\sigma_\infty$: fatigue limit;

γ : geometrical parameter

β : geometrical parameter

Experimental data shown by green dots in fig is fitted by the Stüssi line and above equation is obtained.



(c) α – fatigue life at various amplitudes.

Taking Logarithm

$$\log(N) = \frac{1}{\beta} \log\left(\frac{R_m - \Delta\sigma}{\Delta\sigma - \Delta\sigma_\infty}\right) - \frac{1}{\beta} \log(\gamma)$$

$$Y = AX + B$$

where,

$$Y = \log(N)$$

$$X = \log\left(\frac{R_m - \Delta\sigma}{\Delta\sigma - \Delta\sigma_\infty}\right)$$

$$A = \frac{1}{\beta}$$

$$B = -\frac{1}{\beta} \log(\gamma)$$

$$N_f = kA_0 + \frac{ka}{m \times \sqrt{\pi / 2}} \times e^{\left(-2 \times \left(\frac{\alpha - q}{m}\right)^2\right)}$$

$$A_0 = 1171328$$

$$k = 2.2$$

$$m = 6.37$$

$$q = 7.83$$

$$a = 3.14$$

It's a peak (bell-shaped) function centered around $\alpha = 8 \mu\text{m}$
Nf increases up to $\alpha = 8 \mu\text{m}$, then decreases

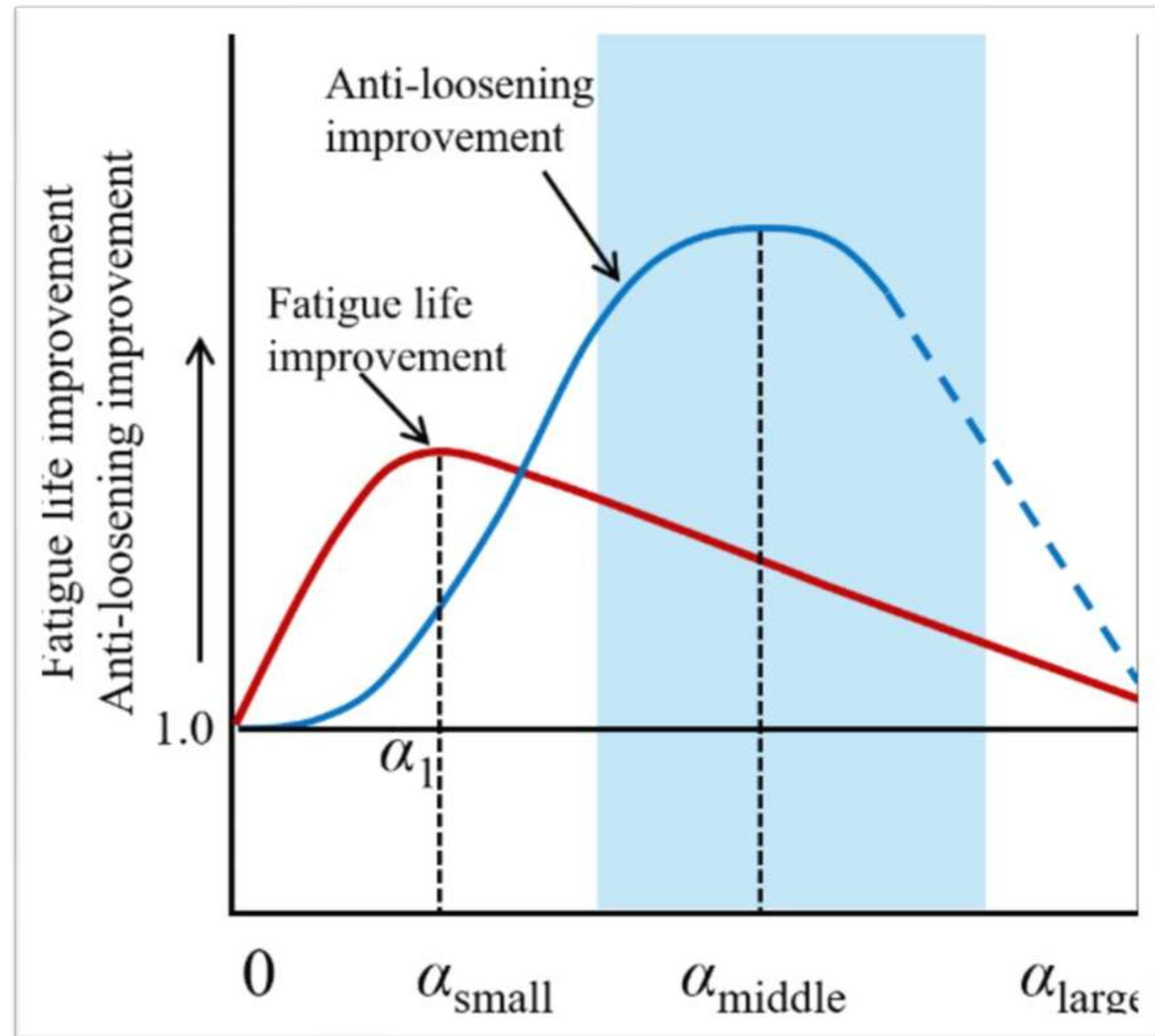
Loosening experiment

$$F_R = B_1\alpha + B_2\alpha^2 + B_3\alpha^3 + B_4\alpha^4 + B_5\alpha^5 + B_6$$

FR = Residual force after vibration

α = Pitch difference between nut and bolt

Results



X-axis: Thread pitch difference (denoted as α)

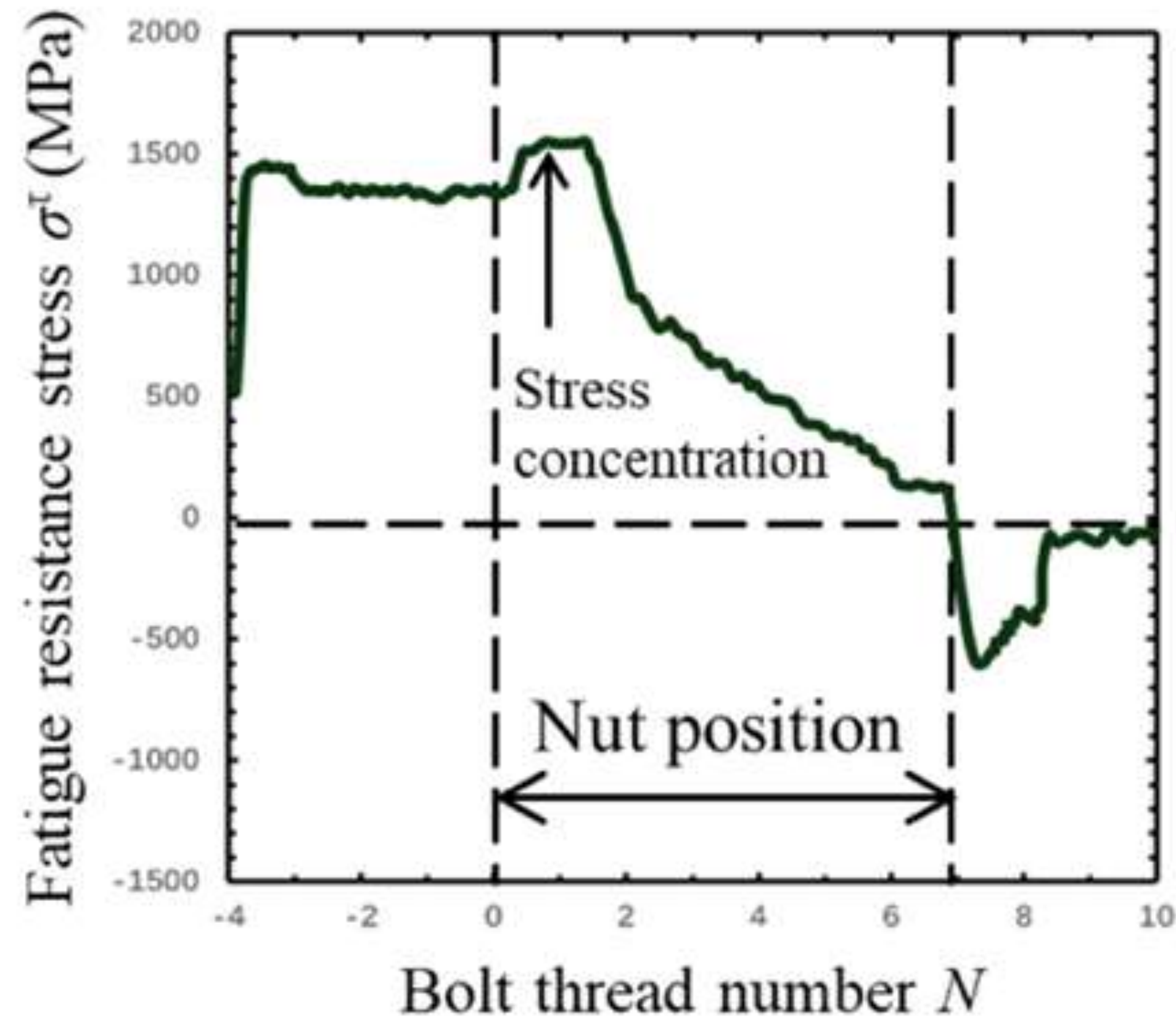
Y-axis: Improvement in

Fatigue Life (blue line)

Anti-Loosening (red line)

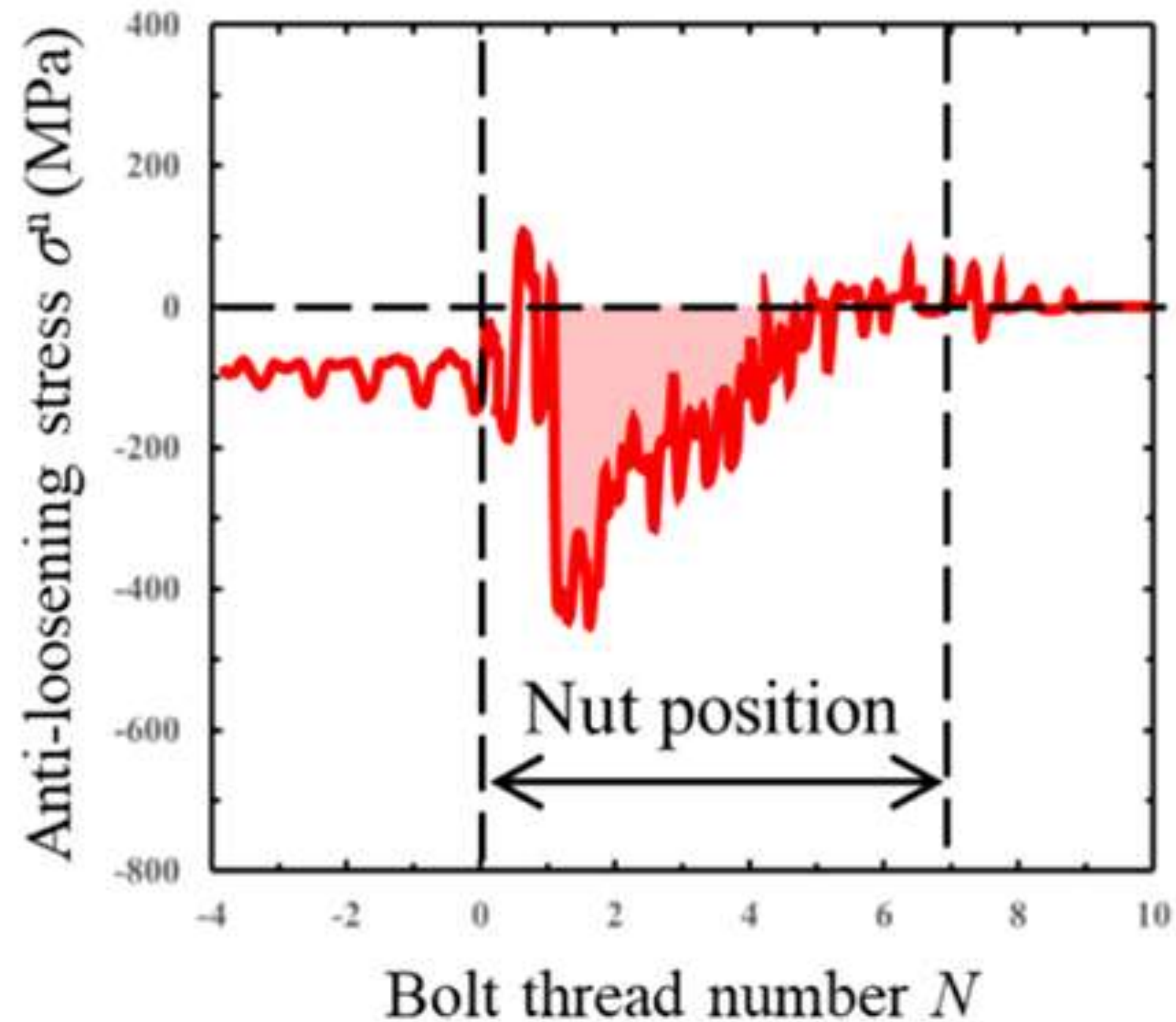
- At α_{small} : High fatigue life, but poor anti-loosening.
- At α_{large} : Great anti-loosening, but poor fatigue life.
- At α_{middle} : Balanced performance

Results

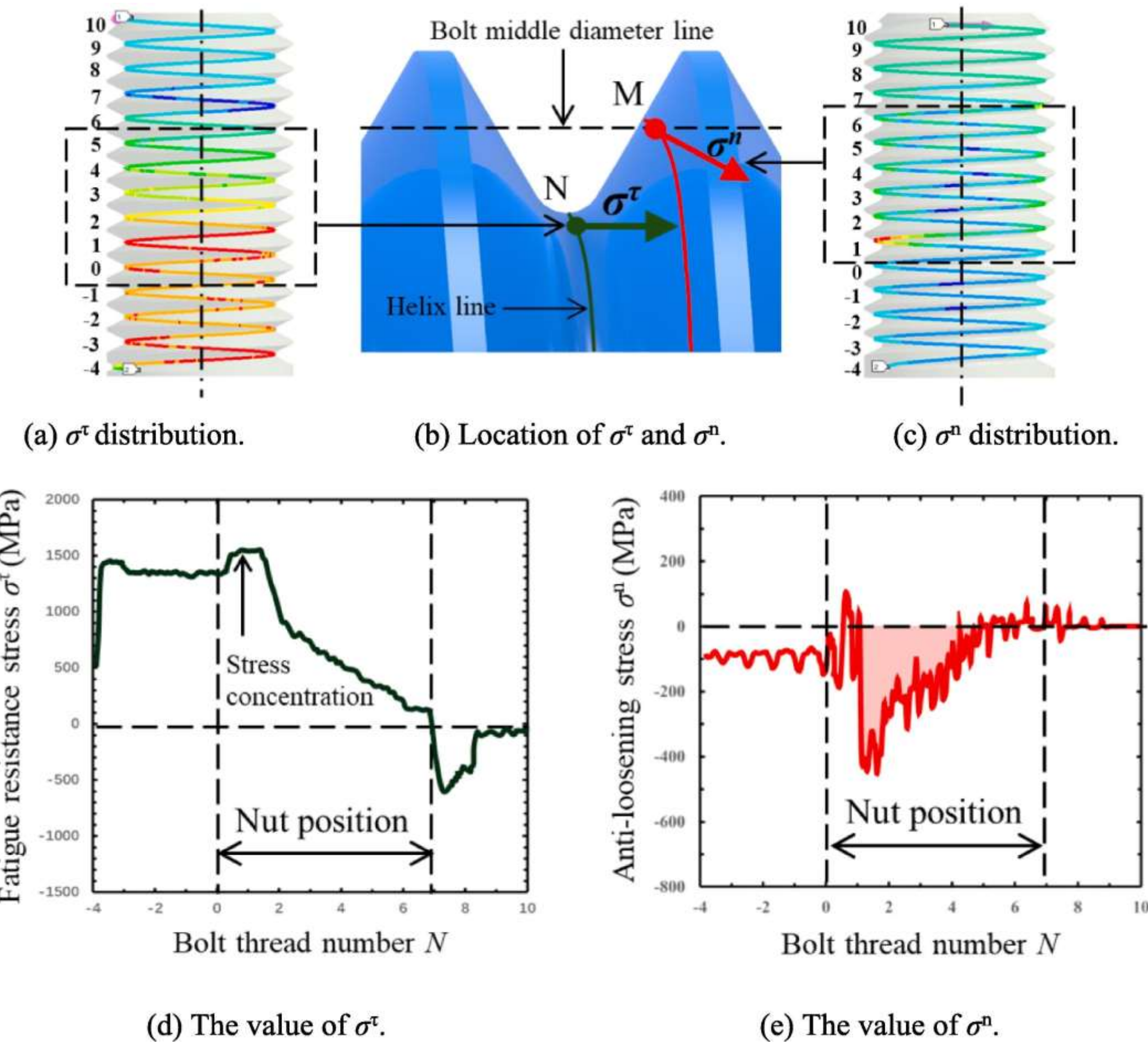


- X-axis: Bolt thread number (N)
- Y-axis: Axial stress (σ^T) in MPa
- The first engaged thread has the highest stress (σ^T), showing a clear stress concentration.
- Stress concentration at the first thread is the key reason for fatigue crack initiation.

Results

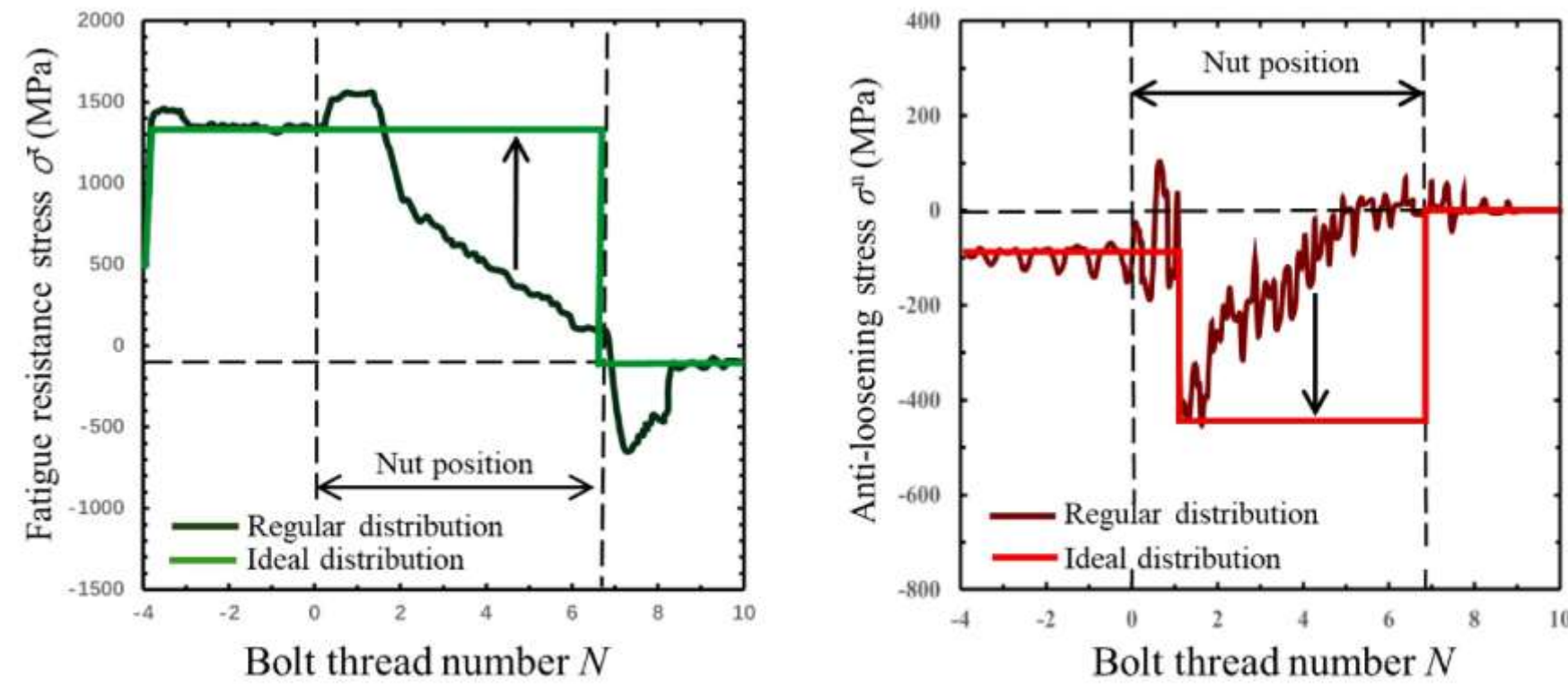


- X-axis: Thread number
- Y-axis: Compressive contact stress (σ_n) in MPa
- Maximum contact stress is also concentrated at the first thread.
- The area under the σ_n curve is directly proportional to total frictional force available to resist loosening.



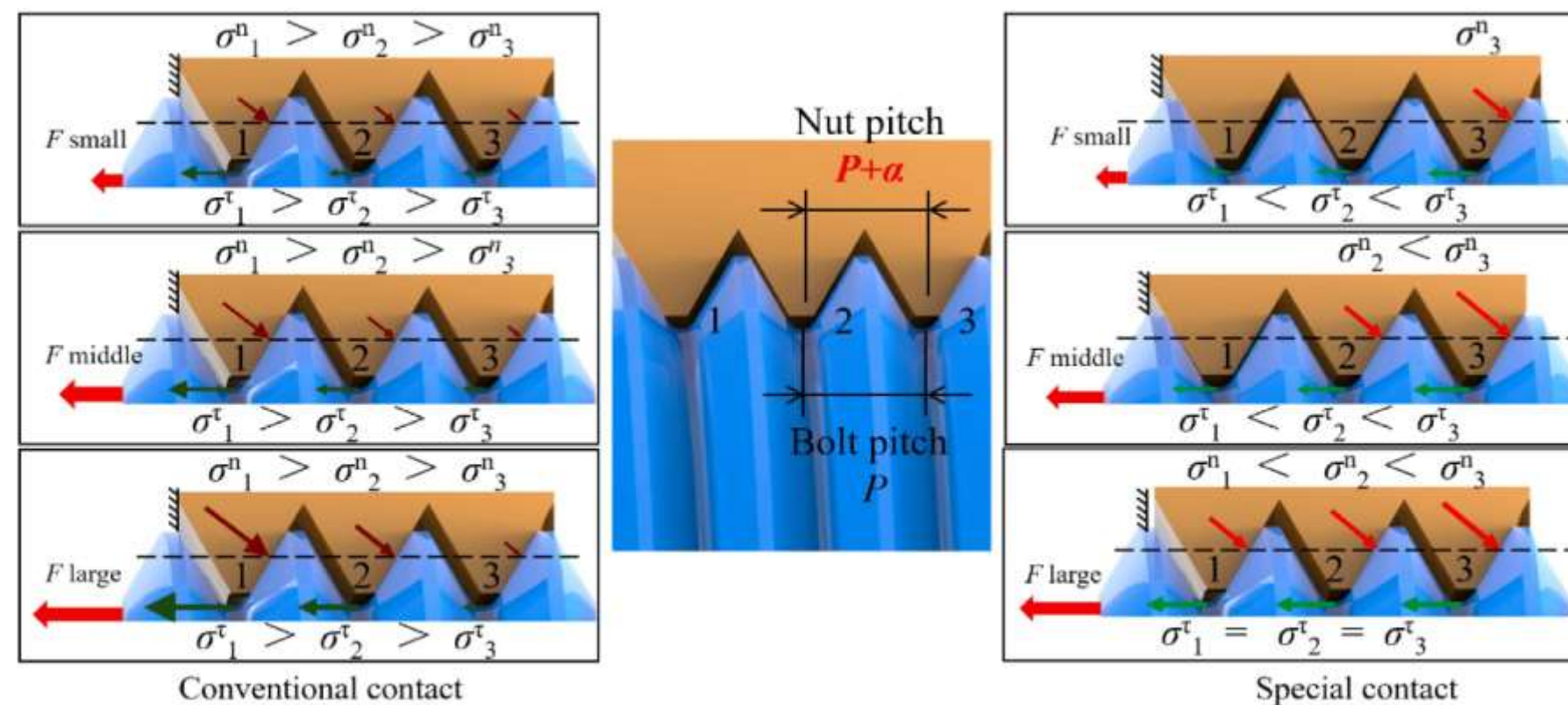
Sub-figure	Description
a)	σ^τ at the root of the threads (under static tightening force $F = 50\% \sigma_s$).
b)	Measurement positions for σ^τ (green arrow) and contact stress (σ_n) (red arrow) on thread surface
c)	Compressive contact stress (σ_n) on thread surfaces under the same preload.
d)	Line plot of σ^τ vs. thread number $\rightarrow$ peak at 1st thread.
e)	Line plot of σ_n vs. thread number $\rightarrow$ also peak at 1st thread.

Results



(a) Ideal fatigue resistance stress.

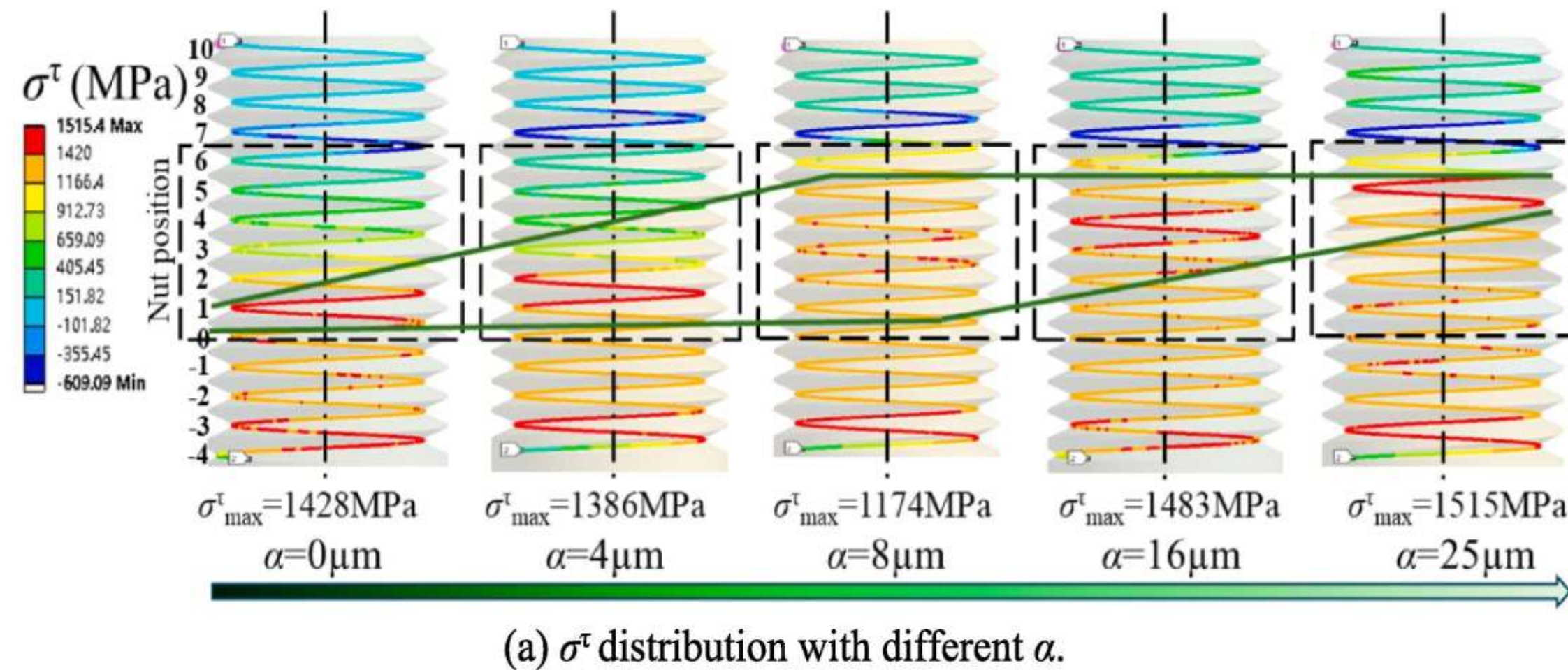
(b) Ideal anti-loosening stress.



(c) Contact status during tightening.

Sub Fig	Description
a)	Comparison of actual (dark green) vs. ideal (light green) fatigue resistance stress (σ^T) distribution across thread turns.
b)	Comparison of actual (red) vs. ideal (light red) anti-loosening stress (σ_n) distribution across thread turns.
c)	A diagram showing how pitch difference affects thread engagement order and enables ideal σ^T and σ_n .

- Current Fasteners Have Bad Stress Concentration:
Both fatigue (σ^T) and loosening (σ_n) issues originate from excessive stress at the first thread.
- Ideal σ^T Distribution:
A uniform axial stress minimizes fatigue crack initiation risk.
- Ideal σ_n Distribution:
A wider contact stress zone leads to better friction and loosening resistance.
Necessary Pitch Difference is required.
- Practical Design :
Use pitch difference as a simple geometric tuning parameter to achieve dual improvement:
Fatigue resistance increase
Anti-loosening increase



- Distribution of σ^τ with change in α .

1. At $\alpha = 0$:

σ^τ is heavily concentrated at the first thread hence high risk of fatigue failure.

2. At $\alpha = 8 \mu\text{m}$:

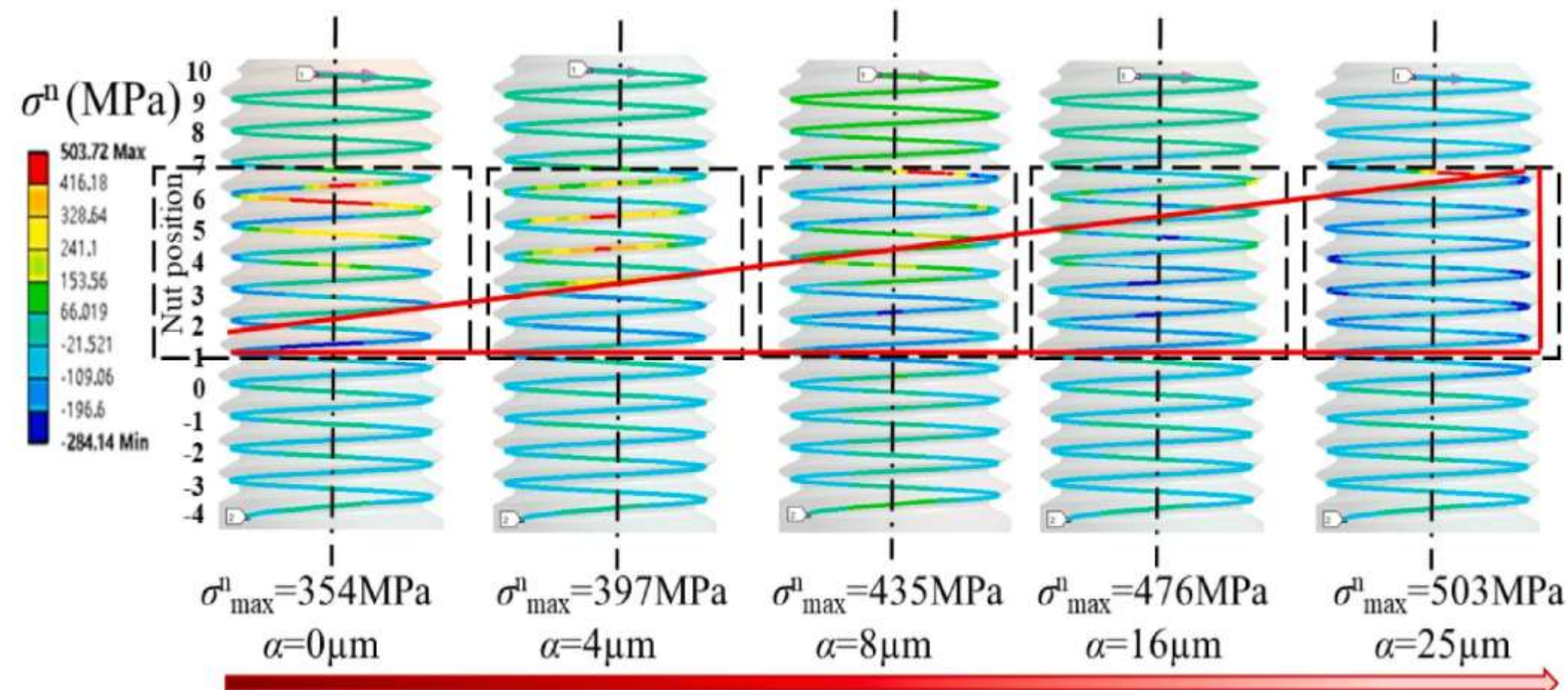
σ^τ is evenly spread across threads.

This is the ideal scenario for fatigue resistance.

3. At $\alpha > 8 \mu\text{m}$:

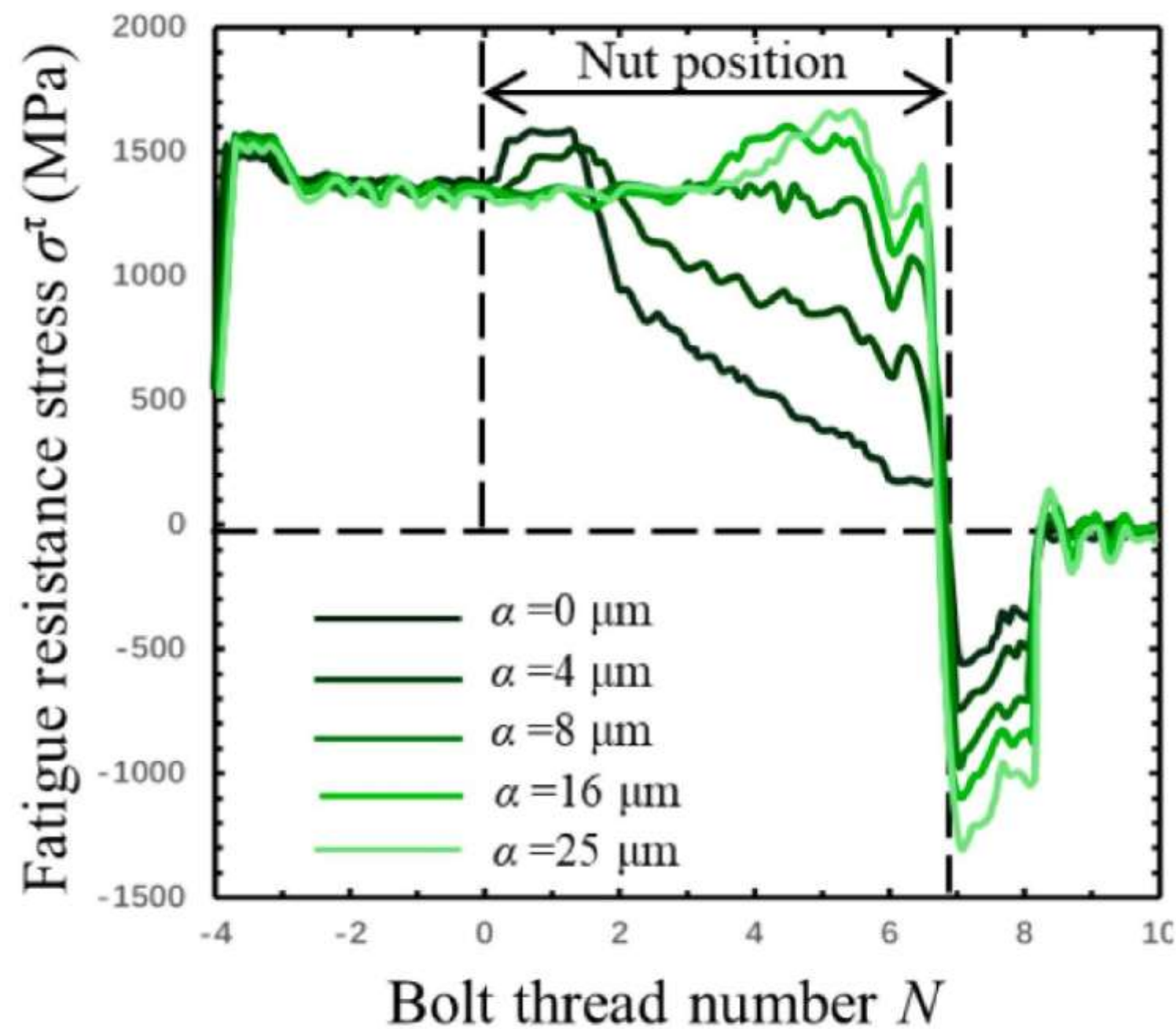
σ^τ begins to concentrate again, but now at higher-numbered threads but can lead to fatigue in another location.

Results



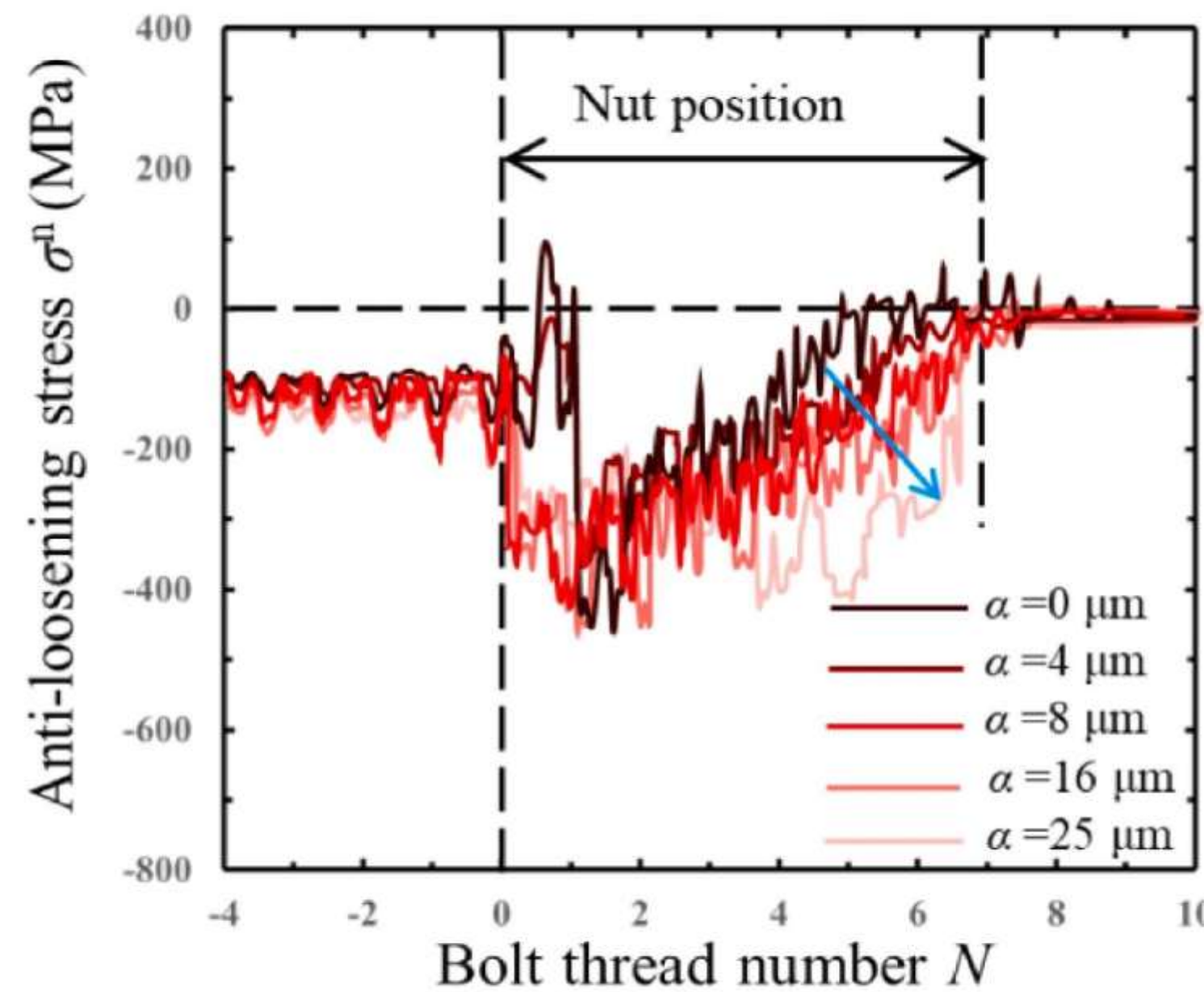
(b) σ_n distribution with different α .

- $\alpha = 0 \mu m$ (no pitch difference):
 σ_n is highly concentrated on the first thread only leads to poor anti-loosening.
 - $\alpha=8 \mu m$:
 σ_n becomes more evenly distributed across multiple threads which causes ideal stress pattern, better friction, stronger anti-loosening.
 - $\alpha > 8 \mu m$
 σ_n covers even more threads causing maximum anti-loosening, but may start affecting fatigue resistance negatively.
-
- σ_n is responsible for anti-loosening performance by generating thread friction.
 - As pitch difference (α) increases, the blue compressive stress area spreads to more threads.
 - The stress shifts from being concentrated on the 1st thread to being shared across all threads.
 - Larger stress area means more thread contact leads to higher friction force which leads to better anti-loosening.



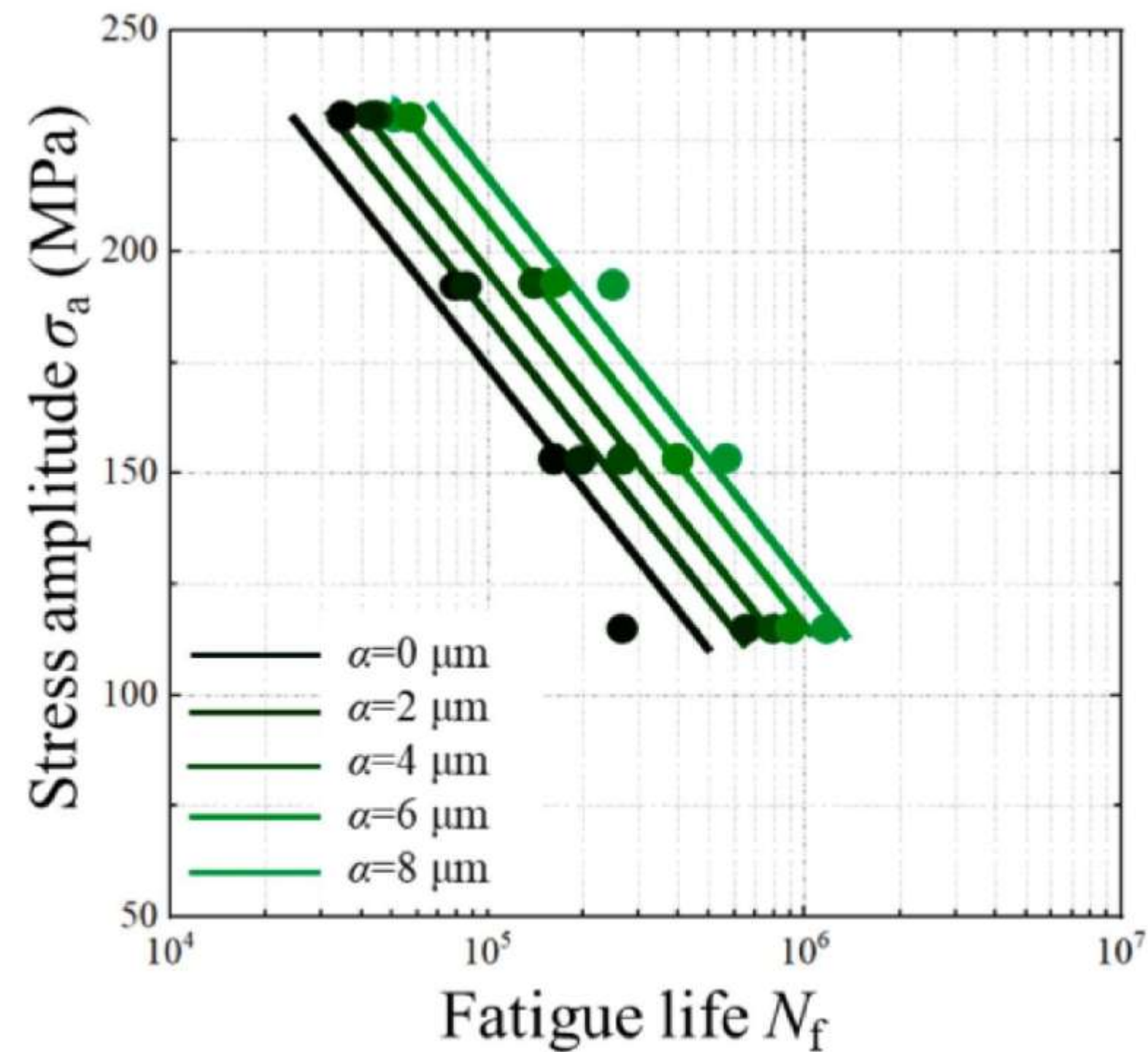
(c) σ^τ value with different α .

- X-axis: Bolt thread number N
- Y-axis: Fatigue resistance stress in σ^τ MPa.
- σ^τ (axial stress at thread root) causes fatigue failure.
- The graph shows σ^τ peak values at different pitch differences (α).
- As α increases from 0 to 8 μm , the peak σ^τ decreases → indicates reduced stress concentration, better for fatigue resistance.
- At $\alpha = 8 \mu\text{m}$, σ^τ reaches its lowest value → optimal for fatigue life.
- For $\alpha > 8 \mu\text{m}$, σ^τ increases again, suggesting stress shifts to higher threads, reducing fatigue performance.



(d) σ^n value with different α .

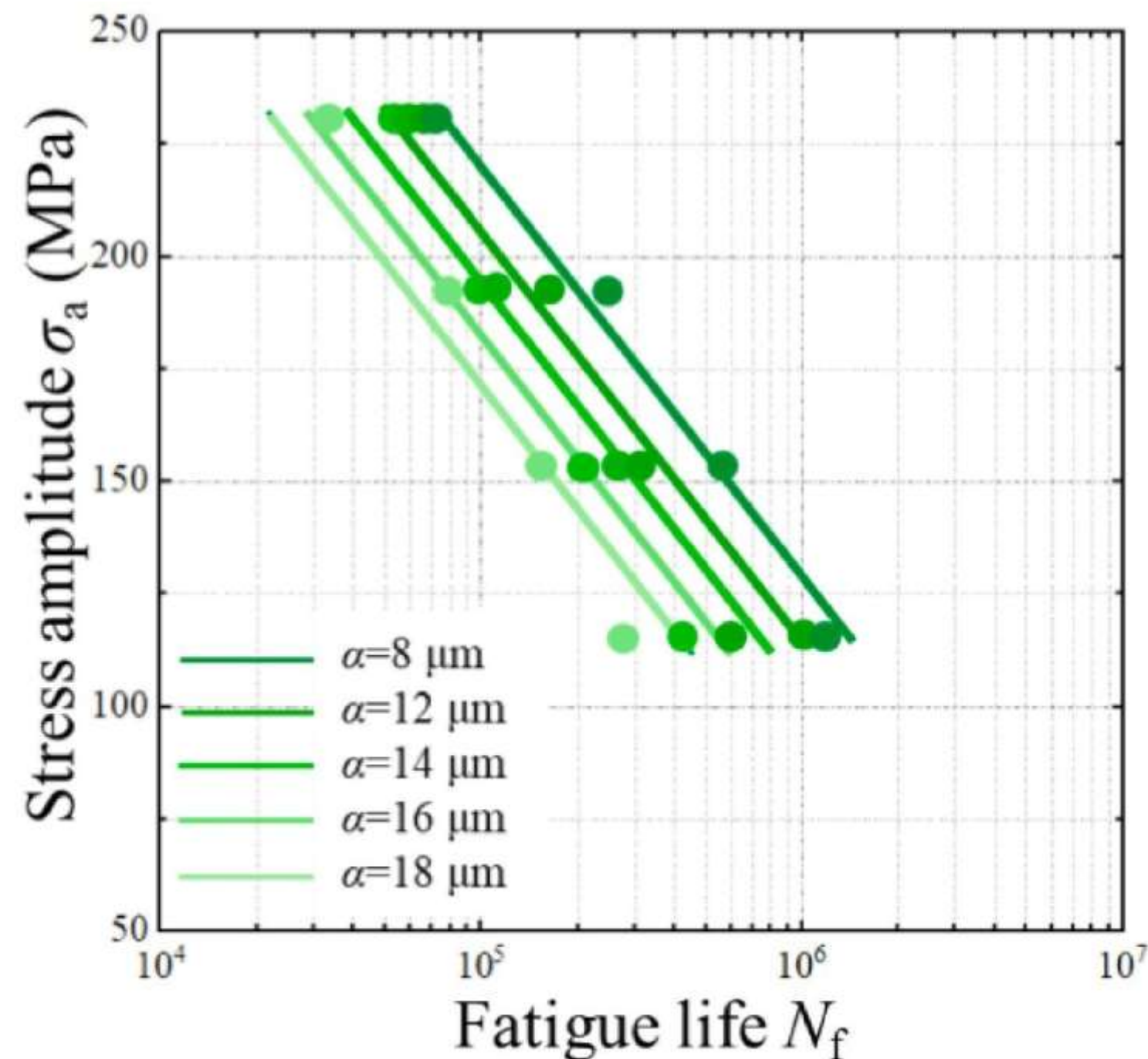
- X-axis: Bolt thread number N
- Y-axis: Anti-loosening stress in σ^n (MPa).
- Shows how the area under σ^n curve (related to contact force) changes with α .
- As α increases, the area under σ^n increases steadily.
- Larger area = more thread contact leads to higher frictional force which leads to better anti-loosening.
- Even beyond $\alpha = 8 \mu\text{m}$, σ^n continues to improve, unlike σ^T .



(a) Number of cycles to fatigue N_f , $\alpha=0\mu\text{m}$ - $8\mu\text{m}$.

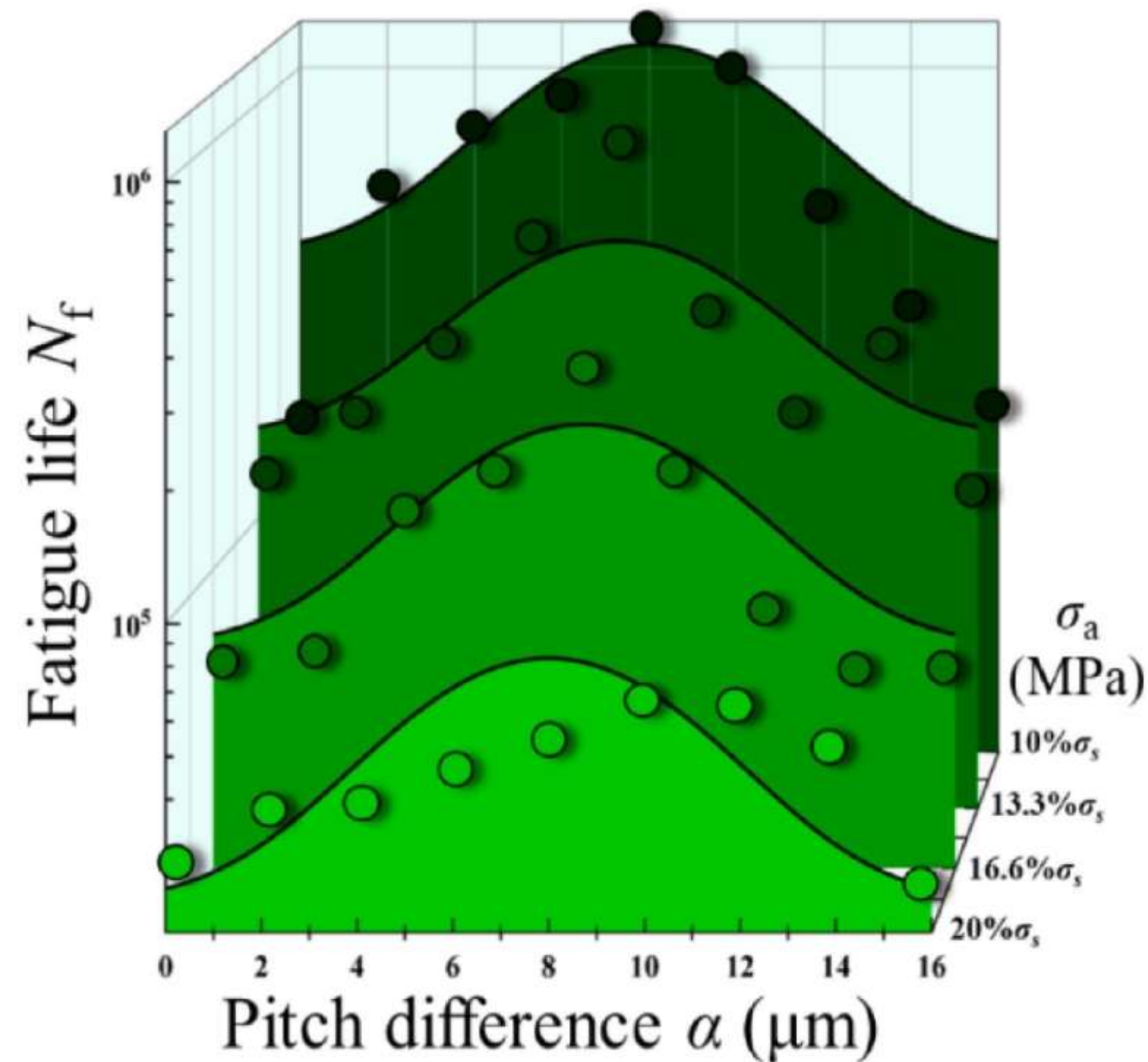
- X-axis: Fatigue life N_f
- Y-axis: Stress amplitude σ_a (MPa).
- S–N curve = Stress amplitude (σ_a) vs. number of cycles to failure (N_f).
- Four curves shown for different pitch differences α : 0, 2, 4, 6, 8 μm .
- As α increases, the curve shifts upward — indicating higher fatigue life (N_f) at all stress amplitudes.
- $\alpha = 8 \mu\text{m}$ fastener lasts the longest under cyclic loading.
- Key insight: A small pitch difference helps distribute stress better, reducing σ_a , improving fatigue performance.

Results



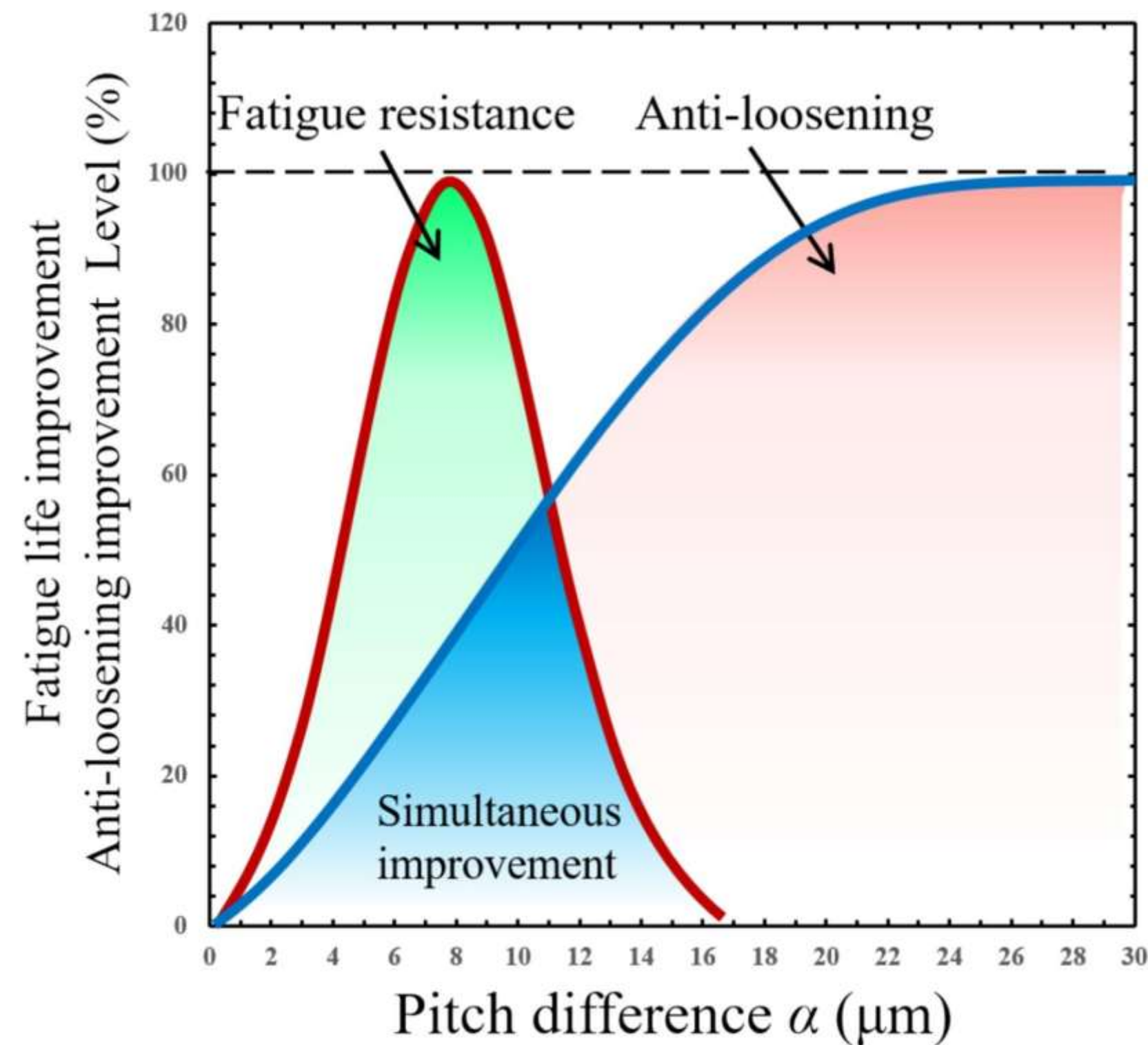
(b) Number of cycles to fatigue N_f , $\alpha = 8 \mu\text{m}$ - $16 \mu\text{m}$.

- X-axis: Fatigue life N_f
- Y-axis: Stress amplitude σ_a (MPa).
- Same setup, now for larger pitch differences: $\alpha = 8, 10, 12, 14, 16 \mu\text{m}$.
- As α increases beyond $8 \mu\text{m}$, the curve shifts downward — fatigue life (N_f) decreases.
- Over-correction leads to new stress concentrations at other threads (not 1st thread), worsening fatigue.
- Optimal fatigue resistance occurs at $\alpha = 8 \mu\text{m}$.
- Shows that increasing α too much reverses the benefit.



(c) α – fatigue life at various amplitudes.

- X-axis: Fatigue life N_f
- Y-axis: Stress amplitude σ_a (MPa).
- Plots fatigue life (N_f) vs. α , for all four stress amplitudes.
- Each curve shows a peak at $\alpha \approx 8 \mu\text{m}$.
- After $\alpha = 8 \mu\text{m}$, N_f drops for all loading amplitudes.
- Statistical curve fitting confirms that fatigue life follows a normal distribution trend w.r.t. α .
- Key conclusion:
 $\alpha = 8 \mu\text{m}$ is the optimal pitch difference for maximizing fatigue life, regardless of stress amplitude.



- **At $\alpha = 0 \mu\text{m}$:**

Fatigue performance is low (stress concentrated at first thread).
Anti-loosening performance is also low (low friction/contact force).

- **As α increases (0-8 μm):**

Fatigue resistance increases sharply.
Anti-loosening performance increases more gradually.

- **At $\alpha \approx 8 \mu\text{m}$:**

Fatigue resistance is maximum.
Anti-loosening continues to rise, but not maximum.
This is the optimum point for fatigue performance.

- **For $\alpha > 8 \mu\text{m}$:**

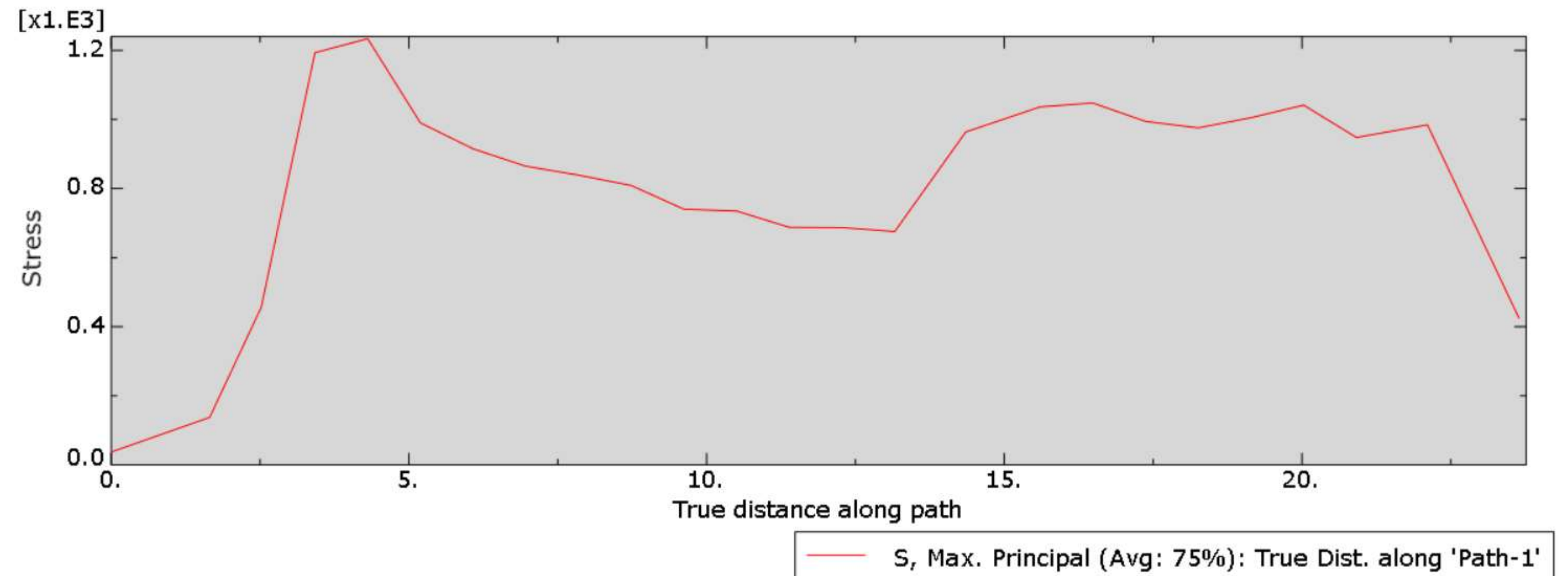
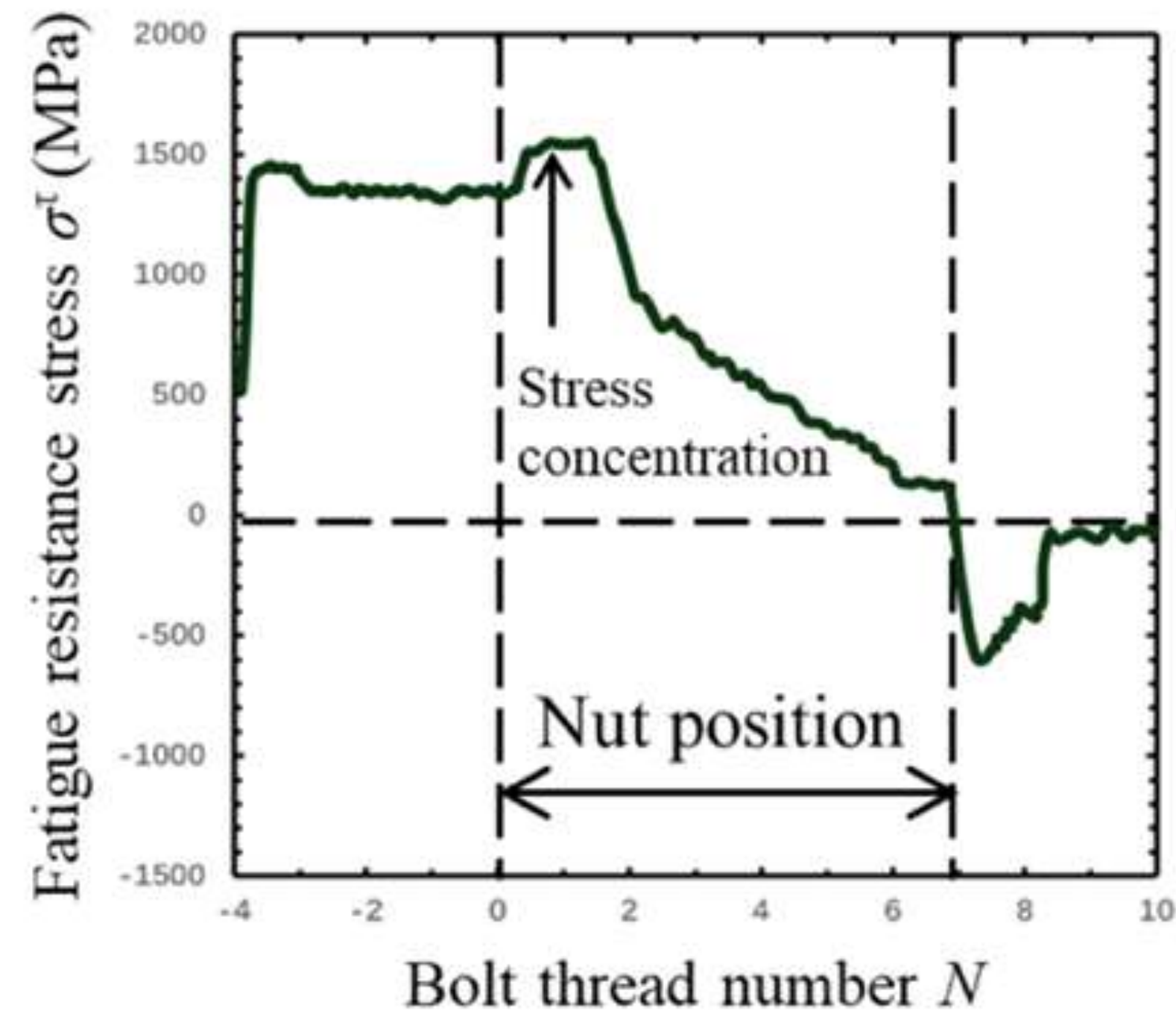
Fatigue performance drops (stress shifts to other threads).
Anti-loosening performance continues to improve, max near $\alpha = 25 \mu\text{m}$.

- **Intersection area (blue shaded region):**

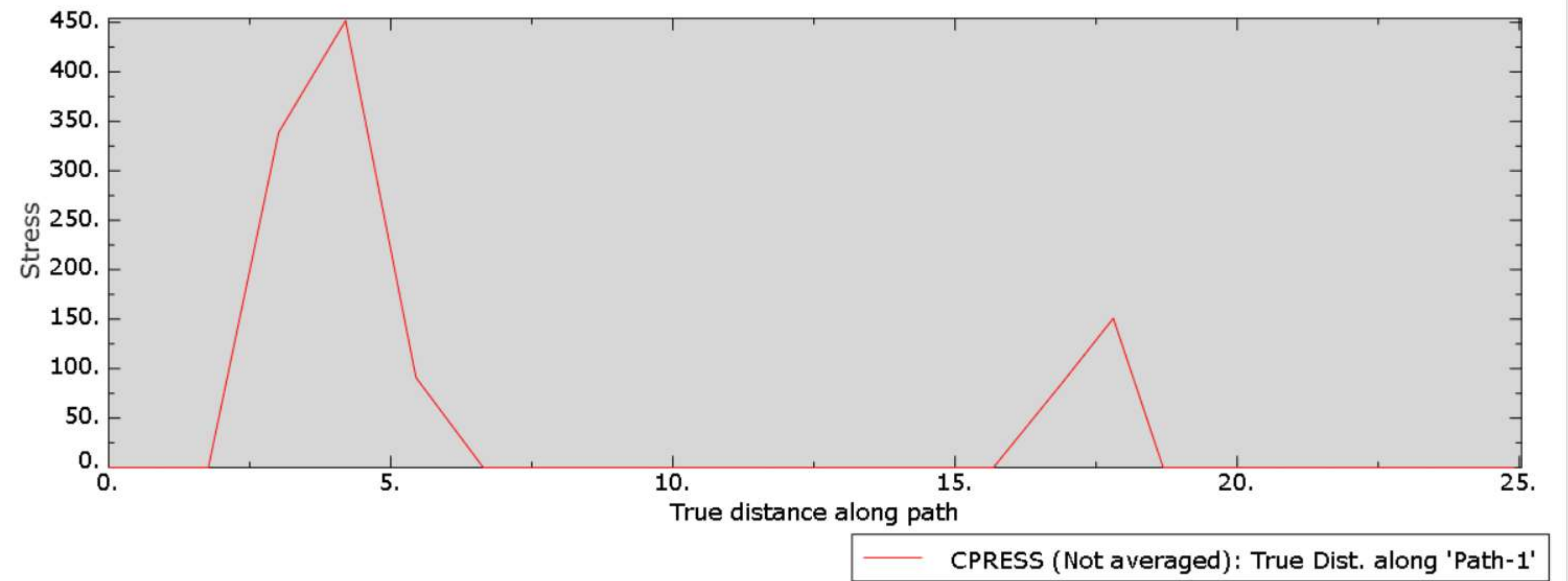
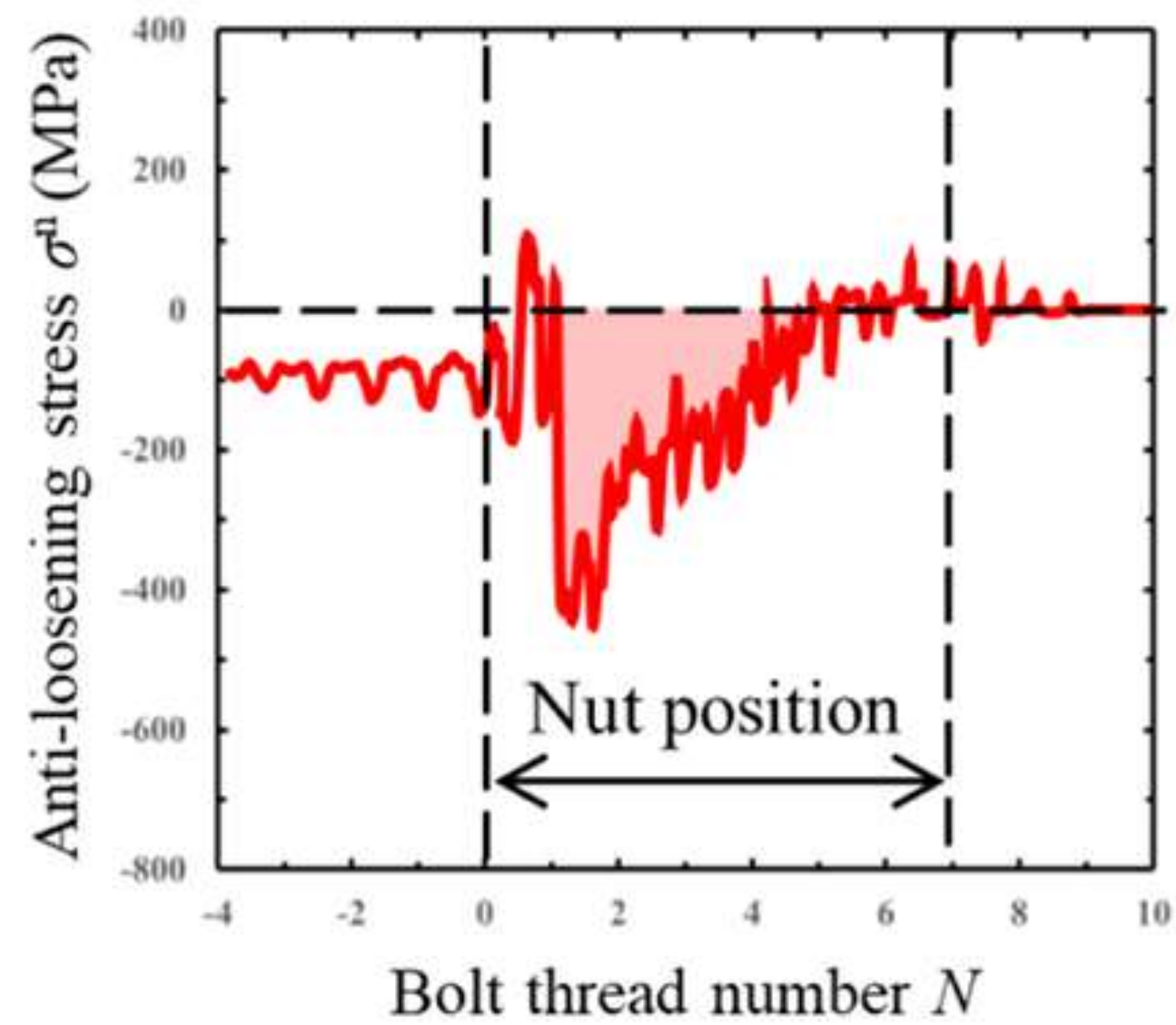
Both fatigue and anti-loosening performance are acceptable.

Results

Mesh size=1 , linear element, Plastic



Results



- A stress-guided design approach is proposed, where fastener performance is optimized by controlling local stress distribution rather than changing material or surface properties.
- Finite element results show that fatigue (σ^T) and anti-loosening (σ_n) are both governed by stress concentration at the first engaged thread.
- Introducing a controlled thread pitch difference (α) deliberately redistributes axial and contact stresses across multiple threads.
- An optimal pitch difference of $\alpha \approx 8 \mu\text{m}$ yields uniform axial stress, maximum fatigue life, and significantly improved friction-based anti-loosening.
- The proposed method offers a simple, cost-effective, geometry-based design framework for achieving simultaneous improvement in fatigue resistance and loosening performance.

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Thank You!

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